MSC 201 **42506**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Outcomes 0 and 10 have probabilities 0.9 and 0.1. Why is their expected value not 5?	A. The values cannot be averaged. B. The outcomes require unequal probability weights. C. The larger value is impossible. D. Expected value always equals the most likely outcome. 10 pts
2. Which expression computes P(X>0) for a nonnegative count X?	A. 1-P(X=0). B. P(X=0). C. 1-P(X=1). D. The mean of X. 10 pts
3. A rare-event binomial count has n=1,000 and p=0.002. What mean would its Poisson approximation use?	A. 0.002. B. 500. C. 1,000. D. 2. 10 pts
4. A fair six-sided die is rolled. What is the probability of a result greater than 4?	A. 5/6. B. 1/6. C. 2/6. D. 4/6. 10 pts
5. Two distinct terminal paths describe exactly one head in two tosses. How is their combined probability found?	A. Subtract the smaller from the larger. B. Multiply the two path probabilities. C. Ignore one because both describe one head. D. Add the two path probabilities. 10 pts
6. How do X>=3 and X>3 differ?	A. X>3 includes first success on trial 3. B. X>=3 requires four initial failures. C. They are the same event. D. X>=3 includes first success on trial 3. 10 pts
7. An independent fair coin has just produced five tails. What is the chance of heads next?	A. Greater than 1/2 because heads is due. B. 1. C. Less than 1/2 because tails is favored. D. 1/2. 10 pts

Test continues on the other side of the sheet.

MSC 201 **42506**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. There are 20 independent trials with p=0.3. What is the expected success count?	A. 20. B. 6. C. 0.3. D. 14. 10 pts
9. Four cables are sampled uniformly without replacement from 12, of which 5 are faulty. Which model directly fits the faulty count?	A. Hypergeometric. B. Geometric waiting count. C. Continuous uniform. D. Normal individual measurement. 10 pts
10. For two trials with positive chance of two failures, how do exactly one failure and at least one failure compare?	A. They must be equal. B. At least one has greater probability. C. The two events are complements. D. Exactly one has greater probability. 10 pts

42506

TEST PAGE 1

1.

B.

10 pts

2. A.

10 pts

3.

D.

10 pts

4.

C.

10 pts

5.

D.

10 pts

6.

D.

10 pts

7.

D.

10 pts

TEST PAGE 2

8.

B.

10 pts

9. A.

10 pts

10.

B.

10 pts

MSC 201 **48680**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A box has 3 red and 2 blue tokens. After a red token is drawn and not replaced, what happens to the red chance on the next draw?	A. It stays 3/5. B. It falls from 3/5 to 2/4. C. It rises from 3/5 to 3/4. D. It becomes zero. 10 pts
2. Among 20 tablet users, 12 also use laptops. What is P(laptop given tablet) in this sample?	A. 12/20. B. 12 divided by all college students. C. 20/12. D. 8/20. 10 pts
3. X counts the successes. Which values satisfy fewer than 3 successes?	A. 1, 2, 3 only. B. 0, 1, 2, 3. C. 3 and above. D. 0, 1, 2. 10 pts
4. A game has negative expected net gain. Can a player still win money on one play?	A. No; the largest prize must be negative. B. Only after the expected value changes. C. Yes, if positive-gain outcomes have positive probability. D. No; every outcome must lose money. 10 pts
5. A box has 8 red and 2 blue tokens; 5 are drawn without replacement. What is the minimum red count?	A. 5. B. 2. C. 3. D. 0. 10 pts
6. A probability table totals 1 but contains one entry of -0.1. What follows?	A. It is valid because only the total matters. B. Negative probabilities simply indicate rare events. C. It is invalid because probabilities cannot be negative. D. It is valid if the mean is positive. 10 pts
7. Which variable is a count rather than a waiting time?	A. Exact time between two requests. B. Duration of a request. C. Requests received in the next five minutes. D. Minutes until the next request. 10 pts

Test continues on the other side of the sheet.

MSC 201 **48680**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. To find the probability of A followed by B when the stages may be dependent, which expression fits?	A. P(A) times P(B given A). B. P(A) plus P(B). C. P(A) minus P(B given A). D. P(A) times P(B) without checking independence. 10 pts
9. X counts successes in 8 trials. Which proposed outcome is impossible?	A. X=4. B. X=0. C. X=8. D. X=9. 10 pts
10. X counts trials through first success. What does X>3 require?	A. A success on trial 3. B. Exactly three successes. C. Failures on the first three trials. D. At least one success in the first three trials. 10 pts

48680

TEST PAGE 1

1.

B.

10 pts

2. A.

10 pts

3.

D.

10 pts

4.

C.

10 pts

5.

C.

10 pts

6.

C.

10 pts

7.

C.

10 pts

TEST PAGE 2

8. A.

10 pts

9.

D.

10 pts

10.

C.

10 pts

MSC 201 **69348**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. P(A)=0.30 and P(A given B)=0.30, with P(B)>0. What does this support?	A. A is impossible. B. B guarantees A. C. A and B are independent. D. A and B are mutually exclusive. 10 pts
2. A binomial count has mean 6. What does this imply about the next experiment?	A. Its count must be exactly 6. B. Its count may differ from 6. C. Its count cannot exceed 6. D. Its count cannot be below 6. 10 pts
3. Which event is the complement of at least one failure in three trials?	A. Exactly one failure. B. At least two failures. C. No failures. D. All three failures. 10 pts
4. A box has 8 red and 2 blue tokens; 5 are drawn without replacement. What is the minimum red count?	A. 0. B. 5. C. 2. D. 3. 10 pts
5. The mean waiting count is 4 trials. Must the first success occur by trial 4?	A. Yes; 4 is the maximum. B. No; longer waits remain possible. C. Yes; the mean is a deadline. D. No; it must occur exactly on trial 5. 10 pts
6. Expected jobs per session are 1.5. What total is expected over 20 such sessions?	A. Exactly 30 in every collection of sessions. B. 1.5. C. 30. D. 21.5. 10 pts
7. X counts heads in three independent fair-coin tosses. Are 0, 1, 2, and 3 heads equally likely?	A. Yes; every fair-coin count is equally likely. B. No; 0 is impossible. C. Yes; there are four possible counts. D. No; 1 and 2 each occur through more sequences. 10 pts

Test continues on the other side of the sheet.

MSC 201 **69348**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A table contains 5 students using both devices. What does this show about mutual exclusivity?	A. They must be independent. B. Neither event has positive probability. C. They are mutually exclusive. D. The two use events are not mutually exclusive. 10 pts
9. A Poisson count has mean 9. What is its standard deviation?	A. 81. B. 9. C. 0. D. 3. 10 pts
10. On a die, A is even and B is greater than 3. Which results satisfy at least one condition?	A. 1, 3 only. B. 2, 4, 5, 6. C. All six results. D. 4, 6 only. 10 pts

69348

TEST PAGE 1

1.

C.

10 pts

2.

B.

10 pts

3.

C.

10 pts

4.

D.

10 pts

5.

B.

10 pts

6.

C.

10 pts

7.

D.

10 pts

TEST PAGE 2

8.

D.

10 pts

9.

D.

10 pts

10.

B.

10 pts

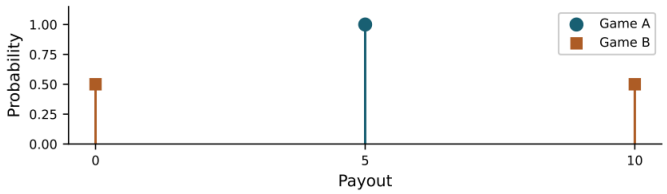
MSC 201 **67780**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A fair coin gives 8 heads in 10 tosses. What follows?	A. The next two tosses must be tails. B. Fairness requires exactly 5 heads in every ten tosses. C. The coin's future head probability is now 0.8. D. Such a short-run imbalance can occur with a fair coin. 10 pts
2. Game A pays 5 with certainty. Game B pays 0 or 10 with equal chance. What is true? 	A. B has a larger expected payout. B. A has greater spread. C. They have equal expected payout; B has greater spread. D. Both have zero spread. 10 pts
3. There are 20 independent trials with p=0.3. What is the expected success count?	A. 20. B. 0.3. C. 14. D. 6. 10 pts
4. X counts failures in exactly 20 independent trials; Y counts arrivals during an hour with no fixed trial total. Which distinction matters?	A. X must be continuous. B. Both must have the same maximum. C. Y must be geometric. D. X fits a binomial setting; Y may fit a Poisson setting if its assumptions hold. 10 pts
5. P(A)=0.30 and P(A given B)=0.30, with P(B)>0. What does this support?	A. A is impossible. B. A and B are mutually exclusive. C. A and B are independent. D. B guarantees A. 10 pts
6. The success chance improves after each failed attempt because the process learns. Why is a geometric model questionable?	A. There is a stopping rule. B. The first success occurs eventually. C. Failures are counted. D. The success probability is not constant. 10 pts

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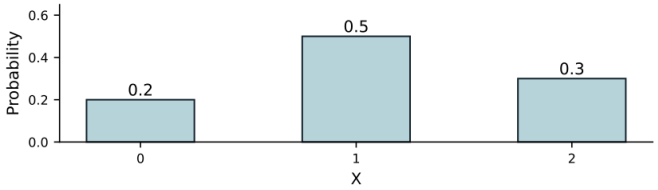
MSC 201 **67780**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. X=0,1,2 has probabilities 0.2,0.5,0.3. What is P(X<2)? 	A. 0.7. B. 1. C. 0.5. D. 0.3. 10 pts
8. A population has 2 faulty cables among 10; 8 are sampled without replacement. What sample information constrains the count?	A. Only the first cable's color. B. Only the population mean. C. Both available faulty cables and available nonfaulty cables. D. Only whether the count is odd. 10 pts
9. Which event is the complement of at least one failure in three trials?	A. All three failures. B. At least two failures. C. No failures. D. Exactly one failure. 10 pts
10. Among 20 tablet users, 12 also use laptops. What is P(laptop given tablet) in this sample?	A. 12/20. B. 12 divided by all college students. C. 20/12. D. 8/20. 10 pts

67780

TEST PAGE 1

1.

D.

10 pts

2.

C.

10 pts

3.

D.

10 pts

4.

D.

10 pts

5.

C.

10 pts

6.

D.

10 pts

TEST PAGE 2

7. A.

10 pts

8.

C.

10 pts

9.

C.

10 pts

10. A.

10 pts

MSC 201 **45575**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A conditioning event has probability zero. What happens to the usual conditional-probability ratio?	A. The ratio automatically equals 0.5. B. The ratio automatically equals 1. C. Zero divided by zero equals zero. D. Its denominator is zero, so this ratio is undefined. 10 pts
2. A constant-rate Poisson process averages 2 requests per minute. What is the mean count over 3 minutes?	A. 6. B. 2/3. C. 3. D. 2. 10 pts
3. Half of tablet users use laptops, and half of non-tablet users use laptops. What does this suggest in these data?	A. Laptop use does not depend on tablet-use status. B. Every student uses exactly one device. C. Every student uses both devices. D. The events are mutually exclusive. 10 pts
4. On a die, A is even and B is greater than 3. Which results satisfy both?	A. 2, 4, 5, 6. B. 4 and 6. C. 2 only. D. 5 only. 10 pts
5. A fixed fee increases by 2 dollars. What happens to net-gain mean and SD?	A. Mean is unchanged; SD rises. B. Mean falls by 2; SD is unchanged. C. Both double. D. Both fall by 2. 10 pts
6. A proposed table gives probabilities 0.2, 0.5, 0.3 for X=0,1,2. Is it valid?	A. No; entries must all be equal. B. Yes; entries are nonnegative and total 1. C. No; X cannot include zero. D. No; one entry exceeds 0.25. 10 pts
7. An independent fair coin has just produced five tails. What is the chance of heads next?	A. 1. B. Less than 1/2 because tails is favored. C. Greater than 1/2 because heads is due. D. 1/2. 10 pts

Test continues on the other side of the sheet.

MSC 201 **45575**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. How do X>=3 and X>3 differ?	A. X>3 includes first success on trial 3. B. X>=3 requires four initial failures. C. They are the same event. D. X>=3 includes first success on trial 3. 10 pts
9. A box has 4 red and 6 blue tokens; 3 are drawn without replacement. What red counts are possible?	A. Only 0 or 3. B. 1, 2, 3, 4. C. 0, 1, 2, 3. D. 0 through 6. 10 pts
10. There are 20 independent trials with p=0.3. What is the expected success count?	A. 0.3. B. 20. C. 6. D. 14. 10 pts

45575

TEST PAGE 1

1.

D.

10 pts

2. A.

10 pts

3. A.

10 pts

4.

B.

10 pts

5.

B.

10 pts

6.

B.

10 pts

7.

D.

10 pts

TEST PAGE 2

8.

D.

10 pts

9.

C.

10 pts

10.

C.

10 pts

MSC 201 **72434**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Half of tablet users use laptops, and half of non-tablet users use laptops. What does this suggest in these data?	A. Laptop use does not depend on tablet-use status. B. Every student uses both devices. C. Every student uses exactly one device. D. The events are mutually exclusive. 10 pts
2. A service meets its deadline with probability 0.80. What is the probability of missing it?	A. 1.80. B. 0.20. C. 0.40. D. 0.80. 10 pts
3. Expected jobs per session are 1.5. What total is expected over 20 such sessions?	A. Exactly 30 in every collection of sessions. B. 1.5. C. 21.5. D. 30. 10 pts
4. A box has 4 red and 6 blue tokens. One red is removed. What is the next red probability?	A. 3/9. B. 4/9. C. 4/10. D. 3/10. 10 pts
5. A late-job rate is 10% overall but 30% during storms. What does storm information do?	A. It proves the events are independent. B. It makes lateness impossible outside storms. C. It changes the estimated chance of lateness. D. It establishes that every storm job is late. 10 pts
6. For two trials with positive chance of two failures, how do exactly one failure and at least one failure compare?	A. At least one has greater probability. B. Exactly one has greater probability. C. They must be equal. D. The two events are complements. 10 pts
7. A supplied probability table lists P(X=0),...,P(X=8). To find P(X<=2), which entries are needed?	A. Only the entry for 2. B. The entries for 3 through 8. C. The expected value. D. The entries for 0, 1, and 2. 10 pts

Test continues on the other side of the sheet.

MSC 201 **72434**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. X counts heads in three independent fair-coin tosses. Are 0, 1, 2, and 3 heads equally likely?	A. No; 1 and 2 each occur through more sequences. B. No; 0 is impossible. C. Yes; there are four possible counts. D. Yes; every fair-coin count is equally likely. 10 pts
9. X counts trials through first success; Y counts failures before it. What is the relationship?	A. Y=X+1. B. X=Y. C. X=2Y. D. X=Y+1. 10 pts
10. Which variable is a count rather than a waiting time?	A. Exact time between two requests. B. Minutes until the next request. C. Duration of a request. D. Requests received in the next five minutes. 10 pts

72434

TEST PAGE 1

1. A.

10 pts

2. B.

10 pts

3. D.

10 pts

4. A.

10 pts

5. C.

10 pts

6. A.

10 pts

7. D.

10 pts

TEST PAGE 2

8. A.

10 pts

9. D.

10 pts

10. D.

10 pts

MSC 201 **48340**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. X counts failures in exactly 20 independent trials; Y counts arrivals during an hour with no fixed trial total. Which distinction matters?	A. Y must be geometric. B. X must be continuous. C. Both must have the same maximum. D. X fits a binomial setting; Y may fit a Poisson setting if its assumptions hold. 10 pts
2. Independent trials have p=0.25. What is the expected number of trials through first success?	A. 3. B. 4. C. 0.25. D. 25. 10 pts
3. Probability is shifted from a low outcome to a higher outcome, with other probabilities unchanged. What happens to expected value?	A. It decreases. B. It must become the largest outcome. C. It increases. D. It cannot change. 10 pts
4. Which event is the complement of at least one failure in three trials?	A. At least two failures. B. All three failures. C. No failures. D. Exactly one failure. 10 pts
5. Among 20 tablet users, 12 also use laptops. What is P(laptop given tablet) in this sample?	A. 12 divided by all college students. B. 12/20. C. 20/12. D. 8/20. 10 pts
6. Two mutually exclusive events each have positive probability. Can they be independent?	A. Yes; both have positive probabilities. B. No; one occurring makes the other impossible. C. Yes; disjoint always means independent. D. Only if their probabilities add to 1. 10 pts
7. Four cables are sampled uniformly without replacement from 12, of which 5 are faulty. Which model directly fits the faulty count?	A. Geometric waiting count. B. Continuous uniform. C. Hypergeometric. D. Normal individual measurement. 10 pts

Test continues on the other side of the sheet.

MSC 201 **48340**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A binomial count has mean 6. What does this imply about the next experiment?	A. Its count may differ from 6. B. Its count cannot exceed 6. C. Its count must be exactly 6. D. Its count cannot be below 6. 10 pts
9. A probability table totals 1 but contains one entry of -0.1. What follows?	A. It is invalid because probabilities cannot be negative. B. It is valid because only the total matters. C. Negative probabilities simply indicate rare events. D. It is valid if the mean is positive. 10 pts
10. A spinner has three colors occupying half, one quarter, and one quarter of its area. Why is assigning each color probability 1/3 inappropriate?	A. A spinner cannot have three outcomes. B. Probabilities must be whole numbers. C. The color areas are unequal. D. Every color must have probability 1/2. 10 pts

48340

TEST PAGE 1

1.

D.

10 pts

2.

B.

10 pts

3.

C.

10 pts

4.

C.

10 pts

5.

B.

10 pts

6.

B.

10 pts

7.

C.

10 pts

TEST PAGE 2

8. A.

10 pts

9. A.

10 pts

10.

C.

10 pts

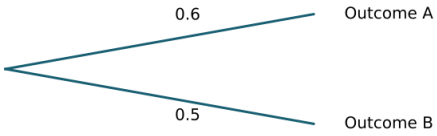
MSC 201 **41495**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. At a tree node, two exhaustive outgoing branch probabilities are 0.6 and 0.5. What is wrong?  	A. They should both be zero. B. All tree branches must be 0.5. C. Branches cannot represent conditional probabilities. D. They total more than 1. 10 pts
2. A probability table totals 1 but contains one entry of -0.1. What follows?	A. Negative probabilities simply indicate rare events. B. It is valid if the mean is positive. C. It is invalid because probabilities cannot be negative. D. It is valid because only the total matters. 10 pts
3. Independent trials have p=0.25. What is the expected number of trials through first success?	A. 0.25. B. 4. C. 25. D. 3. 10 pts
4. A Poisson count has mean 9. What is its standard deviation?	A. 9. B. 0. C. 3. D. 81. 10 pts
5. A box has 4 red and 6 blue tokens; 3 are drawn without replacement. What red counts are possible?	A. Only 0 or 3. B. 1, 2, 3, 4. C. 0 through 6. D. 0, 1, 2, 3. 10 pts
6. For two trials with positive chance of two failures, how do exactly one failure and at least one failure compare?	A. They must be equal. B. The two events are complements. C. Exactly one has greater probability. D. At least one has greater probability. 10 pts

Test continues on the other side of the sheet.

MSC 201 **41495**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Probability is shifted from a low outcome to a higher outcome, with other probabilities unchanged. What happens to expected value?	A. It cannot change. B. It decreases. C. It increases. D. It must become the largest outcome. 10 pts
8. 20 students use both devices; 40 use tablets and 80 use laptops. Which conditional probability is larger?	A. Neither can use the overlap count. B. Laptop use given tablet use. C. Tablet use given laptop use. D. They must be equal. 10 pts
9. Six tokens are drawn without replacement from a small mixed box. What challenges an exact binomial model?	A. The count is an integer. B. The number of draws is fixed. C. There are only two colors. D. Draws change the composition and are dependent. 10 pts
10. On a die, A is even and B is greater than 3. Which results satisfy both?	A. 2, 4, 5, 6. B. 4 and 6. C. 5 only. D. 2 only. 10 pts

41495

TEST PAGE 1

1.

D.

10 pts

2.

C.

10 pts

3.

B.

10 pts

4.

C.

10 pts

5.

D.

10 pts

6.

D.

10 pts

TEST PAGE 2

7.

C.

10 pts

8.

B.

10 pts

9.

D.

10 pts

10.

B.

10 pts

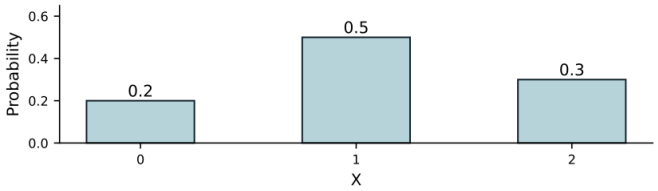
MSC 201 **84014**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. There are 20 independent trials with p=0.3. What is the expected success count?	A. 6. B. 20. C. 14. D. 0.3. 10 pts
2. A first branch has probability 0.4; its next conditional branch is 0.5. What is the path probability?	A. 0.9. B. 0.1. C. 0.8. D. 0.2. 10 pts
3. P(A)=0.4, P(B)=0.5, and their overlap is 0.2. What is P(A OR B)?	A. 0.7. B. 0.2. C. 1.1. D. 0.9. 10 pts
4. Independent trials have p=0.25. What is the expected number of trials through first success?	A. 0.25. B. 3. C. 25. D. 4. 10 pts
5. X=0,1,2 has probabilities 0.2,0.5,0.3. What is P(X<2)? 	A. 0.3. B. 0.7. C. 1. D. 0.5. 10 pts
6. A population is half faulty cables. A random sample of 6 is drawn without replacement. What faulty count is expected?	A. 0.5. B. 3. C. Exactly 3 in every sample. D. 6. 10 pts

Test continues on the other side of the sheet.

MSC 201 **84014**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A count variable has expected value 0.4. What does that mean?	A. The model is invalid. B. Exactly 40% of sessions must contain one event. C. Its long-run average can be 0.4 despite integer observations. D. A single session must contain 0.4 events. 10 pts
8. P(A)=0.30 and P(A given B)=0.30, with P(B)>0. What does this support?	A. B guarantees A. B. A and B are mutually exclusive. C. A and B are independent. D. A is impossible. 10 pts
9. Under a constant errors-per-page Poisson model, document length doubles. What happens to the mean error count?	A. It doubles. B. It halves. C. It becomes the square root of the original mean. D. It stays unchanged. 10 pts
10. A fair six-sided die is rolled. What is the probability of a result greater than 4?	A. 4/6. B. 2/6. C. 1/6. D. 5/6. 10 pts

84014

TEST PAGE 1

1. A.

10 pts

2. D.

10 pts

3. A.

10 pts

4. D.

10 pts

5. B.

10 pts

6. B.

10 pts

TEST PAGE 2

7. C.

10 pts

8. C.

10 pts

9. A.

10 pts

10. B.

10 pts

MSC 201 **94495**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A Poisson count has mean 9. What is its standard deviation?	A. 3. B. 9. C. 81. D. 0. 10 pts
2. To find P(tablet given laptop) from a table, which denominator is needed?	A. The number of laptop users. B. Only the number using neither. C. Only the number using both. D. The number of tablet users. 10 pts
3. A model allows counts 0,1,2,... without a finite upper limit. Can it still be discrete?	A. No; a count must stop at 10. B. Only if every count has equal probability. C. Yes; countably many separated values can be discrete. D. No; all infinite possibilities are continuous. 10 pts
4. A population has 2 faulty cables among 10; 8 are sampled without replacement. What sample information constrains the count?	A. Both available faulty cables and available nonfaulty cables. B. Only whether the count is odd. C. Only the first cable's color. D. Only the population mean. 10 pts
5. The mean waiting count is 4 trials. Must the first success occur by trial 4?	A. Yes; the mean is a deadline. B. No; it must occur exactly on trial 5. C. No; longer waits remain possible. D. Yes; 4 is the maximum. 10 pts
6. Two successes in four trials can occur in several orders. Why must all those orders be considered?	A. The count event includes every order with two successes. B. Only the first successful order can occur. C. Different orders are the same elementary outcome. D. The expected count is an order. 10 pts

Test continues on the other side of the sheet.

MSC 201 **94495**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A service meets its deadline with probability 0.80. What is the probability of missing it?	A. 0.20. B. 0.80. C. 1.80. D. 0.40. 10 pts
8. P(A)=0.30 and P(A given B)=0.30, with P(B)>0. What does this support?	A. A is impossible. B. A and B are independent. C. A and B are mutually exclusive. D. B guarantees A. 10 pts
9. Given A, the chance of B is 0.7. What is the chance of not B given A?	A. 0.7. B. 1.7. C. 0.3. D. Cannot be found unless A is independent of B. 10 pts
10. Outcomes 0 and 10 have probabilities 0.9 and 0.1. Why is their expected value not 5?	A. Expected value always equals the most likely outcome. B. The outcomes require unequal probability weights. C. The values cannot be averaged. D. The larger value is impossible. 10 pts

94495

TEST PAGE 1

1. A.

10 pts

2. A.

10 pts

3. C.

10 pts

4. A.

10 pts

5. C.

10 pts

6. A.

10 pts

TEST PAGE 2

7. A.

10 pts

8. B.

10 pts

9. C.

10 pts

10. B.

10 pts

MSC 201 **32180**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A probability table totals 1 but contains one entry of -0.1. What follows?	A. It is invalid because probabilities cannot be negative. B. It is valid because only the total matters. C. It is valid if the mean is positive. D. Negative probabilities simply indicate rare events. 10 pts
2. A box has 3 red and 2 blue tokens. After a red token is drawn and not replaced, what happens to the red chance on the next draw?	A. It becomes zero. B. It falls from 3/5 to 2/4. C. It rises from 3/5 to 3/4. D. It stays 3/5. 10 pts
3. For two trials with positive chance of two failures, how do exactly one failure and at least one failure compare?	A. They must be equal. B. The two events are complements. C. Exactly one has greater probability. D. At least one has greater probability. 10 pts
4. A first branch has probability 0.4; its next conditional branch is 0.5. What is the path probability?	A. 0.1. B. 0.2. C. 0.9. D. 0.8. 10 pts
5. A box has 4 red and 6 blue tokens. One red is removed. What is the next red probability?	A. 3/9. B. 4/10. C. 4/9. D. 3/10. 10 pts
6. An event is described as drawing a heart or a king. Does the king of hearts qualify?	A. Only if the card is drawn twice. B. No; OR always excludes overlap. C. Yes; the statistical OR includes the overlap. D. Only if hearts and kings are independent. 10 pts
7. Independent trials have p=0.25. What is the expected number of trials through first success?	A. 0.25. B. 25. C. 3. D. 4. 10 pts

Test continues on the other side of the sheet.

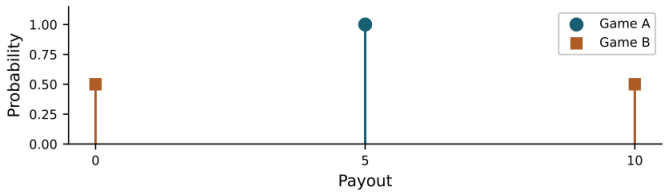
MSC 201 **32180**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A supplied probability table lists P(X=0),...,P(X=8). To find P(X<=2), which entries are needed?	A. The expected value. B. Only the entry for 2. C. The entries for 3 through 8. D. The entries for 0, 1, and 2. 10 pts
9. Game A pays 5 with certainty. Game B pays 0 or 10 with equal chance. What is true? 	A. A has greater spread. B. B has a larger expected payout. C. They have equal expected payout; B has greater spread. D. Both have zero spread. 10 pts
10. Which variable is a count rather than a waiting time?	A. Duration of a request. B. Exact time between two requests. C. Minutes until the next request. D. Requests received in the next five minutes. 10 pts

32180

TEST PAGE 1

1. A.

10 pts

2. B.

10 pts

3. D.

10 pts

4. B.

10 pts

5. A.

10 pts

6. C.

10 pts

7. D.

10 pts

TEST PAGE 2

8. D.

10 pts

9. C.

10 pts

10. D.

10 pts

MSC 201 **50195**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Among 200 recorded jobs, 30 were late. What does 30/200 describe?	A. The exact long-run probability under every possible condition. B. The number of late jobs that must occur tomorrow. C. The observed late-job relative frequency. D. A guarantee that the next job is late. 10 pts
2. X counts trials through first success. What does X>3 require?	A. At least one success in the first three trials. B. A success on trial 3. C. Exactly three successes. D. Failures on the first three trials. 10 pts
3. Which shortcut finds at least one success in a fixed set of independent trials?	A. The expected success count. B. 1 minus the probability of no successes. C. The probability of exactly one success only. D. 1 minus the probability all succeed. 10 pts
4. In many sessions, 20% have no failures. How might this inform a model?	A. Infer that every fifth session must have no failures. B. Use 0.20 as an empirical estimate for P(X=0). C. Set every count probability to 0.20. D. Declare zero failures impossible. 10 pts
5. P(A)=0.4, P(B)=0.5, and their overlap is 0.2. What is P(A OR B)?	A. 0.9. B. 0.7. C. 1.1. D. 0.2. 10 pts
6. A game pays a prize of 8 and costs 3 to play. What is the net gain on that outcome?	A. 11. B. 5. C. 8. D. 3. 10 pts
7. Half of tablet users use laptops, and half of non-tablet users use laptops. What does this suggest in these data?	A. Every student uses exactly one device. B. Every student uses both devices. C. Laptop use does not depend on tablet-use status. D. The events are mutually exclusive. 10 pts

Test continues on the other side of the sheet.

MSC 201 **50195**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Requests average 1 per minute overnight and 8 per minute at noon. What challenges a single constant-rate model for the entire day?	A. Both rates are positive. B. The overnight mean is smaller. C. The rate varies by time of day. D. The events are counts. 10 pts
9. P(A)=0.30 and P(A given B)=0.30, with P(B)>0. What does this support?	A. B guarantees A. B. A and B are independent. C. A and B are mutually exclusive. D. A is impossible. 10 pts
10. A population is half faulty cables. A random sample of 6 is drawn without replacement. What faulty count is expected?	A. 0.5. B. Exactly 3 in every sample. C. 6. D. 3. 10 pts

50195

TEST PAGE 1

1.

C.

10 pts

2.

D.

10 pts

3.

B.

10 pts

4.

B.

10 pts

5.

B.

10 pts

6.

B.

10 pts

7.

C.

10 pts

TEST PAGE 2

8.

C.

10 pts

9.

B.

10 pts

10.

D.

10 pts

MSC 201 **57080**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Game A pays 5 with certainty. Game B pays 0 or 10 with equal chance. What is true? 	A. B has a larger expected payout. B. Both have zero spread. C. A has greater spread. D. They have equal expected payout; B has greater spread. 10 pts
2. P(A)=0.4, P(B)=0.5, and P(A AND B)=0.2. Which check supports independence?	A. 0.4 plus 0.5 equals 0.9. B. 0.4 times 0.5 equals 0.2. C. The intersection is positive. D. 0.5 minus 0.4 equals 0.1. 10 pts
3. X counts heads in three independent fair-coin tosses. Are 0, 1, 2, and 3 heads equally likely?	A. No; 0 is impossible. B. No; 1 and 2 each occur through more sequences. C. Yes; there are four possible counts. D. Yes; every fair-coin count is equally likely. 10 pts
4. After five failures with constant independent p=0.2, what is the chance the next trial succeeds?	A. 1. B. 0.2 divided by 5. C. 0.8. D. 0.2. 10 pts
5. Among 20 tablet users, 12 also use laptops. What is P(laptop given tablet) in this sample?	A. 12/20. B. 12 divided by all college students. C. 8/20. D. 20/12. 10 pts
6. Requests arrive in coordinated bursts. Which Poisson-process assumption deserves scrutiny?	A. Independent arrivals. B. Nonnegative counts. C. The possibility of zero requests. D. Measuring an interval. 10 pts

Test continues on the other side of the sheet.

MSC 201 **57080**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Which shortcut finds at least one success in a fixed set of independent trials?	A. The expected success count. B. 1 minus the probability of no successes. C. The probability of exactly one success only. D. 1 minus the probability all succeed. 10 pts
8. P(A OR B)=0.75. What is the probability of neither?	A. 0.75. B. 1.75. C. 0.50. D. 0.25. 10 pts
9. A fair coin gives 8 heads in 10 tosses. What follows?	A. The coin's future head probability is now 0.8. B. Such a short-run imbalance can occur with a fair coin. C. Fairness requires exactly 5 heads in every ten tosses. D. The next two tosses must be tails. 10 pts
10. A population has 2 faulty cables among 10; 8 are sampled without replacement. What sample information constrains the count?	A. Both available faulty cables and available nonfaulty cables. B. Only whether the count is odd. C. Only the population mean. D. Only the first cable's color. 10 pts

57080

TEST PAGE 1

1.

D.

10 pts

2.

B.

10 pts

3.

B.

10 pts

4.

D.

10 pts

5. A.

10 pts

6. A.

10 pts

TEST PAGE 2

7.

B.

10 pts

8.

D.

10 pts

9.

B.

10 pts

10. A.

10 pts

MSC 201 **82372**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A supplied probability table lists P(X=0),...,P(X=8). To find P(X<=2), which entries are needed?	A. The entries for 0, 1, and 2. B. Only the entry for 2. C. The entries for 3 through 8. D. The expected value. 10 pts
2. Two mutually exclusive events each have positive probability. Can they be independent?	A. Only if their probabilities add to 1. B. Yes; both have positive probabilities. C. Yes; disjoint always means independent. D. No; one occurring makes the other impossible. 10 pts
3. Which sequence gives the first success on trial 4?	A. Four successes. B. Failure, success, failure, success. C. Success, failure, failure, failure. D. Failure, failure, failure, success. 10 pts
4. Among 20 tablet users, 12 also use laptops. What is P(laptop given tablet) in this sample?	A. 12 divided by all college students. B. 8/20. C. 12/20. D. 20/12. 10 pts
5. On a die, A is even and B is greater than 3. Which results satisfy both?	A. 2 only. B. 2, 4, 5, 6. C. 5 only. D. 4 and 6. 10 pts
6. A rare-event binomial count has n=1,000 and p=0.002. What mean would its Poisson approximation use?	A. 2. B. 0.002. C. 500. D. 1,000. 10 pts
7. Outcomes 0 and 10 have probabilities 0.9 and 0.1. Why is their expected value not 5?	A. The larger value is impossible. B. The outcomes require unequal probability weights. C. The values cannot be averaged. D. Expected value always equals the most likely outcome. 10 pts

Test continues on the other side of the sheet.

MSC 201 **82372**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Which expression computes P(X>0) for a nonnegative count X?	A. P(X=0). B. The mean of X. C. 1-P(X=0). D. 1-P(X=1). 10 pts
9. P(A OR B)=0.75. What is the probability of neither?	A. 0.75. B. 0.50. C. 0.25. D. 1.75. 10 pts
10. Which design permits an exact binomial count model with independent randomized draws?	A. Drawing without replacement until the box is empty. B. Always stopping at first red. C. Replacing and remixing each drawn token. D. Drawing most of a small population without replacement. 10 pts

82372

TEST PAGE 1

1. A.

10 pts

2. D.

10 pts

3. D.

10 pts

4. C.

10 pts

5. D.

10 pts

6. A.

10 pts

7. B.

10 pts

TEST PAGE 2

8. C.

10 pts

9. C.

10 pts

10. C.

10 pts

MSC 201 **60299**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A model allows counts 0,1,2,... without a finite upper limit. Can it still be discrete?	A. Only if every count has equal probability. B. Yes; countably many separated values can be discrete. C. No; all infinite possibilities are continuous. D. No; a count must stop at 10. 10 pts
2. After five failures with constant independent p=0.2, what is the chance the next trial succeeds?	A. 0.8. B. 0.2 divided by 5. C. 0.2. D. 1. 10 pts
3. X counts the successes. Which values satisfy fewer than 3 successes?	A. 1, 2, 3 only. B. 0, 1, 2, 3. C. 0, 1, 2. D. 3 and above. 10 pts
4. A box has 3 red and 7 blue tokens; 5 are drawn without replacement. Why is 4 red impossible?	A. Only 3 red tokens exist. B. Red probability must be zero. C. Half the population is drawn. D. The sample contains 5 tokens. 10 pts
5. Two distinct terminal paths describe exactly one head in two tosses. How is their combined probability found?	A. Ignore one because both describe one head. B. Subtract the smaller from the larger. C. Multiply the two path probabilities. D. Add the two path probabilities. 10 pts
6. A fair die is rolled. A means even; B means 6. Why are these events dependent?	A. They cannot happen together. B. Both are separate die rolls. C. Knowing B guarantees A, changing its probability. D. A fair die makes all events independent. 10 pts
7. Which shortcut finds at least one success in a fixed set of independent trials?	A. The expected success count. B. The probability of exactly one success only. C. 1 minus the probability of no successes. D. 1 minus the probability all succeed. 10 pts

Test continues on the other side of the sheet.

MSC 201 **60299**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A constant-rate Poisson process averages 2 requests per minute. What is the mean count over 3 minutes?	A. 6. B. 3. C. 2/3. D. 2. 10 pts
9. Given A, the chance of B is 0.7. What is the chance of not B given A?	A. Cannot be found unless A is independent of B. B. 1.7. C. 0.7. D. 0.3. 10 pts
10. A game's expected prize is 4 dollars. Which entry fee makes expected net gain zero?	A. 8 dollars. B. 4 dollars. C. The maximum possible prize regardless of its probability. D. 0 dollars. 10 pts

60299

TEST PAGE 1

1.

B.

10 pts

2.

C.

10 pts

3.

C.

10 pts

4. A.

10 pts

5.

D.

10 pts

6.

C.

10 pts

7.

C.

10 pts

TEST PAGE 2

8. A.

10 pts

9.

D.

10 pts

10.

B.

10 pts

MSC 201 **26424**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Which variable is a count rather than a waiting time?	A. Exact time between two requests. B. Requests received in the next five minutes. C. Duration of a request. D. Minutes until the next request. 10 pts
2. Which Venn region is the complement of A OR B?	A. Inside A only. B. Outside both circles. C. Every region inside either circle. D. Only the overlap. 10 pts
3. Which shortcut finds at least one success in a fixed set of independent trials?	A. 1 minus the probability all succeed. B. The expected success count. C. 1 minus the probability of no successes. D. The probability of exactly one success only. 10 pts
4. The sample is an unordered set of cables. Why count combinations rather than different orders of the same set?	A. Combinations assume replacement. B. Selection order determines whether a cable is faulty. C. Every sample set has only one cable. D. Reordering the same selected cables does not create a new sample set. 10 pts
5. Outcomes are failure, failure, success. When X counts trials through first success, what is X?	A. 0. B. 2. C. 3. D. 1. 10 pts
6. Given A, the chance of B is 0.7. What is the chance of not B given A?	A. 0.7. B. Cannot be found unless A is independent of B. C. 1.7. D. 0.3. 10 pts
7. A proposed table gives probabilities 0.2, 0.5, 0.3 for X=0,1,2. Is it valid?	A. No; X cannot include zero. B. No; entries must all be equal. C. No; one entry exceeds 0.25. D. Yes; entries are nonnegative and total 1. 10 pts

Test continues on the other side of the sheet.

MSC 201 **26424**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A fixed fee increases by 2 dollars. What happens to net-gain mean and SD?	A. Both double. B. Mean is unchanged; SD rises. C. Mean falls by 2; SD is unchanged. D. Both fall by 2. 10 pts
9. 20 students use both devices; 40 use tablets and 80 use laptops. Which conditional probability is larger?	A. Laptop use given tablet use. B. They must be equal. C. Tablet use given laptop use. D. Neither can use the overlap count. 10 pts
10. An event is described as drawing a heart or a king. Does the king of hearts qualify?	A. Yes; the statistical OR includes the overlap. B. Only if the card is drawn twice. C. Only if hearts and kings are independent. D. No; OR always excludes overlap. 10 pts

26424

TEST PAGE 1

1.

B.

10 pts

2.

B.

10 pts

3.

C.

10 pts

4.

D.

10 pts

5.

C.

10 pts

6.

D.

10 pts

7.

D.

10 pts

TEST PAGE 2

8.

C.

10 pts

9. A.

10 pts

10. A.

10 pts

MSC 201 **72583**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. There are 20 independent trials with p=0.3. What is the expected success count?	A. 6. B. 0.3. C. 20. D. 14. 10 pts
2. A probability table totals 1 but contains one entry of -0.1. What follows?	A. Negative probabilities simply indicate rare events. B. It is valid because only the total matters. C. It is valid if the mean is positive. D. It is invalid because probabilities cannot be negative. 10 pts
3. A table contains 5 students using both devices. What does this show about mutual exclusivity?	A. The two use events are not mutually exclusive. B. Neither event has positive probability. C. They are mutually exclusive. D. They must be independent. 10 pts
4. A late-job rate is 10% overall but 30% during storms. What does storm information do?	A. It makes lateness impossible outside storms. B. It establishes that every storm job is late. C. It proves the events are independent. D. It changes the estimated chance of lateness. 10 pts
5. A spinner has three colors occupying half, one quarter, and one quarter of its area. Why is assigning each color probability 1/3 inappropriate?	A. Probabilities must be whole numbers. B. Every color must have probability 1/2. C. The color areas are unequal. D. A spinner cannot have three outcomes. 10 pts
6. A fixed fee increases by 2 dollars. What happens to net-gain mean and SD?	A. Both double. B. Both fall by 2. C. Mean is unchanged; SD rises. D. Mean falls by 2; SD is unchanged. 10 pts
7. A rare-event binomial count has n=1,000 and p=0.002. What mean would its Poisson approximation use?	A. 2. B. 0.002. C. 1,000. D. 500. 10 pts

Test continues on the other side of the sheet.

MSC 201 **72583**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Which event is the complement of at least one failure in three trials?	A. At least two failures. B. No failures. C. All three failures. D. Exactly one failure. 10 pts
9. The mean waiting count is 4 trials. Must the first success occur by trial 4?	A. No; longer waits remain possible. B. Yes; 4 is the maximum. C. Yes; the mean is a deadline. D. No; it must occur exactly on trial 5. 10 pts
10. A box has 4 red and 6 blue tokens. One red is removed. What is the next red probability?	A. 4/9. B. 3/9. C. 3/10. D. 4/10. 10 pts

72583

TEST PAGE 1

1. A.

10 pts

2. D.

10 pts

3. A.

10 pts

4. D.

10 pts

5. C.

10 pts

6. D.

10 pts

7. A.

10 pts

TEST PAGE 2

8. B.

10 pts

9. A.

10 pts

10. B.

10 pts

MSC 201 **33404**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. For two trials with positive chance of two failures, how do exactly one failure and at least one failure compare?	A. Exactly one has greater probability. B. At least one has greater probability. C. They must be equal. D. The two events are complements. 10 pts
2. Ten independent trials each have success probability 0.3. X counts successes. What makes a binomial model appropriate?	A. The expected count is an integer. B. Success probability exceeds 0.5. C. Fixed trial count, two outcomes, independence, and constant success chance. D. Every trial must succeed. 10 pts
3. Outcomes 0 and 10 have probabilities 0.9 and 0.1. Why is their expected value not 5?	A. The outcomes require unequal probability weights. B. The larger value is impossible. C. The values cannot be averaged. D. Expected value always equals the most likely outcome. 10 pts
4. The sample is an unordered set of cables. Why count combinations rather than different orders of the same set?	A. Combinations assume replacement. B. Reordering the same selected cables does not create a new sample set. C. Every sample set has only one cable. D. Selection order determines whether a cable is faulty. 10 pts
5. Which quantity naturally has discrete possible values?	A. An unrounded file-transfer duration. B. The number of failed files. C. An unrounded measured length. D. The exact mass of a package. 10 pts
6. A fair six-sided die is rolled. What is the probability of a result greater than 4?	A. 5/6. B. 2/6. C. 4/6. D. 1/6. 10 pts

Test continues on the other side of the sheet.

MSC 201 **33404**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. To find P(tablet given laptop) from a table, which denominator is needed?	A. The number of tablet users. B. The number of laptop users. C. Only the number using both. D. Only the number using neither. 10 pts
8. P(A)=0.4, P(B)=0.5, and P(A AND B)=0.2. Which check supports independence?	A. 0.5 minus 0.4 equals 0.1. B. 0.4 plus 0.5 equals 0.9. C. The intersection is positive. D. 0.4 times 0.5 equals 0.2. 10 pts
9. The mean waiting count is 4 trials. Must the first success occur by trial 4?	A. Yes; 4 is the maximum. B. No; it must occur exactly on trial 5. C. Yes; the mean is a deadline. D. No; longer waits remain possible. 10 pts
10. A rare-event binomial count has n=1,000 and p=0.002. What mean would its Poisson approximation use?	A. 1,000. B. 2. C. 500. D. 0.002. 10 pts

33404

TEST PAGE 1

1.

B.

10 pts

2.

C.

10 pts

3. A.

10 pts

4.

B.

10 pts

5.

B.

10 pts

6.

B.

10 pts

TEST PAGE 2

7.

B.

10 pts

8.

D.

10 pts

9.

D.

10 pts

10.

B.

10 pts

MSC 201 **68976**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Four cables are sampled uniformly without replacement from 12, of which 5 are faulty. Which model directly fits the faulty count?	A. Continuous uniform. B. Geometric waiting count. C. Hypergeometric. D. Normal individual measurement. 10 pts
2. P(A)=0.4, P(B)=0.5, and their overlap is 0.2. What is P(A OR B)?	A. 1.1. B. 0.9. C. 0.7. D. 0.2. 10 pts
3. A Venn summary has 30 in A, including 8 in both A and B. How many are in A only?	A. 38. B. 8. C. 22. D. 30. 10 pts
4. A distribution gives P(X=0)=0.2 and P(X=1)=0.5, with only X=0,1,2 possible. What is P(X=2)?	A. 1.3. B. 0.5. C. 0.3. D. 0.7. 10 pts
5. A late-job rate is 10% overall but 30% during storms. What does storm information do?	A. It changes the estimated chance of lateness. B. It proves the events are independent. C. It establishes that every storm job is late. D. It makes lateness impossible outside storms. 10 pts
6. Which shortcut finds at least one success in a fixed set of independent trials?	A. The probability of exactly one success only. B. The expected success count. C. 1 minus the probability all succeed. D. 1 minus the probability of no successes. 10 pts
7. An event is described as drawing a heart or a king. Does the king of hearts qualify?	A. Yes; the statistical OR includes the overlap. B. Only if the card is drawn twice. C. Only if hearts and kings are independent. D. No; OR always excludes overlap. 10 pts

Test continues on the other side of the sheet.

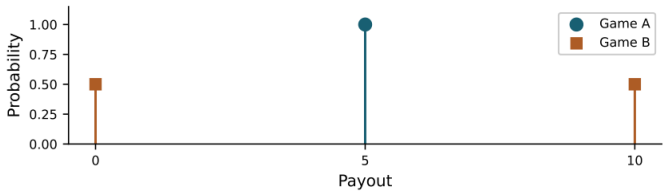
MSC 201 **68976**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A rare-event binomial count has n=1,000 and p=0.002. What mean would its Poisson approximation use?	A. 1,000. B. 2. C. 500. D. 0.002. 10 pts
9. Game A pays 5 with certainty. Game B pays 0 or 10 with equal chance. What is true? 	A. They have equal expected payout; B has greater spread. B. B has a larger expected payout. C. Both have zero spread. D. A has greater spread. 10 pts
10. Which sequence gives the first success on trial 4?	A. Success, failure, failure, failure. B. Four successes. C. Failure, failure, failure, success. D. Failure, success, failure, success. 10 pts

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TEST PAGE 1

1.

C.

10 pts

2.

C.

10 pts

3.

C.

10 pts

4.

C.

10 pts

5. A.

10 pts

6.

D.

10 pts

7. A.

10 pts

TEST PAGE 2

8.

B.

10 pts

9. A.

10 pts

10.

C.

10 pts

MSC 201 **52588**

Unit 2 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A machine's success chance steadily falls as it warms. Which binomial requirement is questionable?	A. Constant success probability across trials. B. A fixed number of trials. C. A numerical success count. D. There being two possible outcomes. 10 pts
2. The success chance improves after each failed attempt because the process learns. Why is a geometric model questionable?	A. There is a stopping rule. B. The first success occurs eventually. C. The success probability is not constant. D. Failures are counted. 10 pts
3. P(A)=0.4, P(B)=0.5, and P(A AND B)=0.2. Which check supports independence?	A. 0.5 minus 0.4 equals 0.1. B. 0.4 plus 0.5 equals 0.9. C. The intersection is positive. D. 0.4 times 0.5 equals 0.2. 10 pts
4. Device-use events appear independent in one class's table. What conclusion needs caution?	A. They are independent for all students at the college. B. The table records that class. C. Their rates match within the observed table. D. The comparison used conditional rates. 10 pts
5. For two trials with positive chance of two failures, how do exactly one failure and at least one failure compare?	A. They must be equal. B. At least one has greater probability. C. Exactly one has greater probability. D. The two events are complements. 10 pts
6. On a die, A is even and B is greater than 3. Which results satisfy both?	A. 2, 4, 5, 6. B. 4 and 6. C. 2 only. D. 5 only. 10 pts
7. Expected jobs per session are 1.5. What total is expected over 20 such sessions?	A. 1.5. B. 21.5. C. 30. D. Exactly 30 in every collection of sessions. 10 pts

Test continues on the other side of the sheet.

MSC 201 **52588**

Unit 2 Test

Name: _____ Date: _____

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Circle one response per question in the right column.

8. A probability table totals 1 but contains one entry of -0.1. What follows?	A. It is valid because only the total matters. B. It is valid if the mean is positive. C. Negative probabilities simply indicate rare events. D. It is invalid because probabilities cannot be negative. 10 pts
9. Ten cables are sampled without replacement from 100,000. Why might binomial be a useful approximation?	A. The small sample barely changes the population composition. B. The faulty count is continuous. C. No-replacement draws become exactly independent. D. All samples have the same faulty count. 10 pts
10. X counts failures in exactly 20 independent trials; Y counts arrivals during an hour with no fixed trial total. Which distinction matters?	A. Both must have the same maximum. B. Y must be geometric. C. X fits a binomial setting; Y may fit a Poisson setting if its assumptions hold. D. X must be continuous. 10 pts

52588

TEST PAGE 1

1. A.

10 pts

2. C.

10 pts

3. D.

10 pts

4. A.

10 pts

5. B.

10 pts

6. B.

10 pts

7. C.

10 pts

TEST PAGE 2

8. D.

10 pts

9. A.

10 pts

10. C.

10 pts